American Heart Association
README file

A README file is often the first file someone will open in your data repository. The purpose of this file is to provide a roadmap for your data deposit and contextualize your project by providing metadata for the inquirer.

Instructions: Fill out this document to its entirety and upload with your data deposit in an American Heart Association approved repository.

Each data deposit into a repository requires a README file and a Data Dictionary.

General information

6/23/2026

1. PI's name: Ryan Roemmich

2. PI's ORCiD ID: https://orcid.org/0000-0003-0797-6455

3. Association award number: 935556

4. Association award doi: Enter Association award doi

5. Association award title: Dynamic treadmill walking: customizable treadmill training for improving gait symmetry after stroke

6. Date range of data collection: 11/7/2022-7/15/2024

7. List of up to six key words to describe data topic:
   a. Gait analysis
   b. walking
   c. Gait asymmetry
   d. Biomechanical phenomena
   e. Cardiopulmonary exercise test
   f. Enter Keyword 6

8. Other: N/A

Data and file overview

1. For each file name, a short description of what data it contains.
   Metabolic Summary Code
   - Code: MetabolicSummary_2026-05-19.Rmd
   - Data: SummaryData_NetPower_2025-03-20.csv
   - Completed code with embedded results: MetabolicSummary_2026-05-19.html (download to view in native format)

   Speed Plotting Code
   - Code: SpeedPlots_2026-05-19.Rmd
   - Data:

    - CBcontrols_TrialAvgSpeeds_2026-03-10.csv - Average speeds across full trial
    - CBcontrols_Last30AvgSpeeds_202603-24.csv - Average speeds across last 30 strides
    - rmse_max_table.csv - Root mean square values for estimated speeds
- Completed code with embedded results: SpeedPlots_2026-05-19.html

Metabolic and Biomechanical Correlations

- Code: MetabolicsBiomechanics_2026-05-19.Rmd
- Data:
    - SummaryData_NetPower_2025-03-20.csv - Net metabolic power data, same as used in Metabolic Summary Code
    - ChangeData_NetPower_2026-05-15.csv - Net metabolic power, formatted as change from baseline 1.125m/s condition
    - DynamicTread_KineticsKinematics_CBcontrols_2025-04-17.csv - Average kinematic and kinetic data
    - DynamicTread_Asymmetries_CBcontrols_2025-04-17.csv - Calculated kinematic and kinetic asymmetries
    - DynamicTread_CBControls_CrossCorrelation_Averages_2026-03-03.xlsx - Cross correlation data
- Completed code with embedded results: MetabolicsBiomechanics_2026-05-19.Rmd

Participant Demographics

- Data: ParticipantDemographics_DynamicTreadmillMetabolics.xlsx

2. Other: N/A

Note the data files you upload should be in a common format, accessible by anyone (Excel, Word, pdf, CSV, txt). Data files should not require a specialized software or application to open. Any analyses performed should be described below in sharing and access information (software, hardware with version).

The Association's Open Science Policy states: Any factual data that is needed for independent verification of research results must be made freely and publicly available in an Association-approved repository as soon as possible, and no later than the time of an associated publication or the end of the award period (and any no-cost extension), whichever comes first.

Sharing and access information

1. Licenses on data (The Association recommends CC-0 + attribution or CC-BY 4.0): MIT License

2. Licenses on source code: MIT License

3. Links to other data deposits/repository links: https://github.com/SKATElin215/DynamicTreadmillMetabolics/

4. Recommended citation for the data: Banks, C., Li, J., Brooke Hall, Stenum, J., & Roemmich, R. (2026). Energy Expenditure During Walking With a Novel Treadmill Controller That Induces Gait Asymmetry. American Society of Neurorehabilitation (ASNR), San Antonio, TX, USA. Zenodo. https://doi.org/10.5281/zenodo.20293358

Methodological information

1. Description of model:
    a. Human, animal: Human
        i. Animal model, enter additional information: N/A
    b. Sex: 5 M/8 F
    c. Sample size: 13
    d. Age: 24+/3, range 19-31
    e. Disease state: Healthy young adults
    f. Interventions: N/A
    g. Comparisons: N/A
    h. Other: N/A

2. Description of methods for data collection (may include links or references to publications or other documents): Text below taken from preprint, https://doi.org/10.64898/2026.05.20.726615

    Study protocol

    Participants walked at three Baseline speeds for five minutes each: 0.75m/s, 1.125m/s, and 1.50m/s (**Figure 1**). Next, they walked with the dynamic treadmill controller engaged for six ten-minute trials. The first and last trials with the controller were open-loop and unconstrained (i.e., we instructed participants simply to walk with the treadmill as it fluctuated between 0.75m/s and 1.50m/s). Participants then completed four *Metronome* trials (*Fast, Slow, Accelerate, Decelerate*) in a randomized order. During these trials, we instructed the participants to time their right heel-strike with the tone of a metronome, which was timed according to that individual's average stride time as measured during the first minute of the 1.125 m/s Baseline trial. This is a replication of the Experiment 2 protocol from our previous study (**Browne *et al*., 2023**), with the addition of metabolic testing.

    The dynamic treadmill controller runs the treadmill via a custom script utilizing the D-Flow (Motek Medical, Amsterdam, NL) programming environment. The treadmill fluctuates between 0.75m/s and 1.50m/s with an acceleration rate of 6.0m/s^2. Within each gait cycle, the controller changes speed at time intervals corresponding to 50% of each individual's stride time from the Baseline 1.125m/s trial. In the *Metronome* trials, the software plays an audible beep to the pace the participant's right heel-strike when the belts are moving at 1.50m/s (*Fast*), 0.75m/s (*Slow*), approximately 1.125m/s while accelerating to the faster speed (*Accelerate*), and approximately 1.125m/s while decelerating to the slower speed (*Decelerate*). Following the six dynamic treadmill controller trials, participants completed another five-minute trial of conventional walking at 1.50m/s for comparison of metabolic data before and after the dynamic treadmill trials.

    Data collection

    Participants walked on an instrumented split-belt treadmill (Motek Medical, Amsterdam, NL) surrounded by a nine-camera three-dimensional motion capture system (Vero, Vicon Motion Systems, Centennial, CO, USA; 100 Hz; RRID: SCR_015001). Participants always wore a

fall-arrest harness that did not provide body weight support when walking on the treadmill. The treadmill has emergency stop buttons located near each study team member, and light gates at the front and back of the treadmill that stop the belts so participants cannot walk off either end. Passive retroreflective markers were placed according to the Vicon lower body Plug-in Gait marker set with the addition of bilateral fifth metatarsal heads, medial malleoli, medial femoral epicondyles, greater trochanters, iliac crest, and trunk markers placed approximately at C7, T10, jugular notch, and xiphoid process (**Kadaba *et al*., 1990**; **Davis *et al*., 1991**; **Vicon Motion Systems, Ltd., 2024**). Participants wore their own walking shoes and fitted athletic clothing. Three-dimensional ground reaction forces were collected at 1000 Hz using the treadmill's built-in force plates under each belt.
Metabolic data were collected using a TrueOne 2400 system (Parvomedics, Sandy, UT, USA). We sampled breath-by-breath oxygen consumption and carbon dioxide production. Baseline metabolic rate was measured during two minutes of quiet standing.

COVID-19 protocol

Most of the data collection for this study was conducted in the fall and winter of 2022-2023. Because expired air from the metabolic chamber was released into the room air, we took additional precautions to lower the risk of exposure to COVID-19 for our participants and study staff. We ran a high efficiency particulate air purifier (Coway Airmega, Seoul, South Korea) on the highest setting throughout the duration of the experiment, placed next to the expired air outlet. Study staff self-screened for respiratory illness symptoms and did not come into the building if they were feeling ill. Staff were fit tested for N-95 masks by the Johns Hopkins Medicine Respiratory Protection Program and wore them throughout the data collection.

Participants took a Flowflex COVID-19 rapid antigen test (ACON Laboratories, San Diego, CA, USA) after being consented and prior to removing their facemask and being set up for metabolic testing. Finally, 'Do Not Enter – Metabolic Experiment in Progress," signs were placed on lab doors to minimize traffic and contact with non-study personnel. No participants declined to follow our procedures, and no participants tested positive on the day of study enrollment.

3. Description of methods used for processing data (how were data processed from raw data to analyzed data): Text below taken from preprint, https://doi.org/10.64898/2026.05.20.726615

Data analysis

We used custom MATLAB (The Mathworks, Natick, MA, USA; RRID: SCR_001622) software for data analysis. We filtered marker data using a fourth order low-pass Butterworth filter with a cut-off frequency of 6 Hz. We determined heel-strike and toe-off events as the timings of the maximum and minimum values of the limb angle trajectory (defined as the sagittal angle between a vector drawn from the iliac crest marker to the second metatarsal head marker and a vector from the iliac crest to the ground). We calculated spatiotemporal and kinematic variables for comparison to our previous work and correlation with metabolic power. Information on how these metrics are calculated can be found in our previous publication (**Browne *et al*., 2023**). Average values from the last 30 strides of each trial are included in all statistical analyses.

Due to difficulties with treadmill software recording belt speeds, we had to calculate belt speed using marker data for six participants. To confirm accuracy of our calculation method, we calculated speeds for the remaining trials and compared to the recorded belt speeds, yielding peak speed root mean square errors of 0.03±0.04m/s.

We calculated metabolic power using Brockway's equation derived for energy expenditure. We then calculated net metabolic power by subtracting baseline metabolic power (from a static standing trial) and normalizing to each participant's body mass. Finally, we calculated summary values over the last two minutes of each gait trial once participants had reached a metabolic steady state. Technical difficulties with the treadmill computer required excluding the final minute of *Decelerate* trial for one participant: accordingly, we calculated metabolic power over minutes 7-9 for this participant.

Statistical analysis

We conducted all statistical analyses in R version 4.3.1 (**R Core Team, 2023**; RRID: SCR_001905) via RStudio version 2023.06.1 (**Posit Team, 2024**; RRID: SCR_001905). We utilized the following R packages for all analyses: tidyverse (RRID: SCR_019186), readxl (RRID: SCR_018083), ggplot2 (RRID:SCR_014601), plyr (RRID:SCR_026985), svglite, desctools (RRID:SCR_027454), and dplyr (RRID:SCR_016708) (**Wickham, 2011**, **2016**; **Wickham & Bryan, 2019**; Wickham *et al*., 2019, 2023, 2025; **Signorell, 2025**). We assessed net metabolic power data for normality using the Shapiro-Wilk W test and data were normally distributed (W=0.98, p=0.19). We also tested net powers for homogeneity of variance using Bartlett's test (K^2=4.53, df=9, p=0.87). We utilized repeated measures ANOVA with a within-subjects factor of Trial to compare metabolic power across the different walking conditions using the rstatix package (**Kassambara, 2021**). We then conducted post-hoc pairwise comparisons using two Dunnett's tests, where we compared net metabolic power during the dynamic treadmill conditions against the metabolic power measured by normal walking at the average of the two dynamic treadmill belt speeds (1.125m/s) and at the fast belt speed (1.50m/s). For validation purposes, we also performed a repeated measures ANOVA on the three Baseline speeds alone, compared pre-post differences in the 1.50m/s conditions using a two-tailed t-test, and repeated our repeated measures ANOVA using block order rather than grouping by dynamic treadmill condition to assess potential order effects. We then assessed the association between net metabolic power and spatiotemporal, kinematic, and kinetic measures of gait symmetry using linear regression. Because the open loop condition did not show consistent changes in gait symmetry in the initial study, we maintained it in the study protocol but do not focus our results on the open loop conditions. We set α≤0.05 for all analyses.

4. Describe any software or instruments needed to interpret or understand the data/results of your study. Please be specific to include software and hardware version numbers. Data files in .xlsx or .csv format. Compiled data output in .html format. Statistical analyses, contained in .Rmd files, conducted in R version 4.3.1 (R Core Team, 2023; RRID: SCR_001905) via RStudio version 2023.06.1 (Posit Team, 2024; RRID: SCR_001905). We utilized the following R packages for all analyses: tidyverse (RRID: SCR_019186), readxl (RRID: SCR_018083), ggplot2 (RRID:SCR_014601), plyr (RRID:SCR_026985), svglite, desctools (RRID:SCR_027454), and dplyr (RRID:SCR_016708) (Wickham, 2011, 2016; Wickham & Bryan, 2019; Wickham et al., 2019, 2023, 2025; Signorell, 2025).

5. Other: N/A